Cody Slater

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About

MD student and PhD candidate working where biological and machine intelligence meet. I think the best way to understand a complex system is to learn to control it. I combine machine learning models with transcriptomic engineering to design molecular tools that measure and control cell-type-specific perturbations at scale. Each experiment retrains the models that co-design the next.

Education

Doctor of Medicine — New York, NY
Columbia University — Expected 2027

Doctor of Philosophy, Biomedical Engineering — New York, NY
Columbia University — Expected 2027

Master of Philosophy, Biomedical Engineering — New York, NY
Columbia University — Expected 2026

Master of Science, Biomedical Engineering — New York, NY
Columbia University — May 2023

Bachelor of Science, Aerospace Engineering — Worcester, MA
Worcester Polytechnic Institute — May 2016

Research & Teaching Experience

Vicković Technology Innovation Lab, New York Genome Center, Graduate Researcher (Ph.D.) — New York, NY
deep learning · PyTorch · GPU HPC · single-cell & spatial transcriptomics · lentivirus · AAV · Cas9 · Cas13 — 2023 – present

- Leading development of large specialized models for RNA modification prediction and generative design.
- Established the first selection-based functional assay for RNA modifications; 10,000+ construct pools.
- Engineering AAV vectors for cell-type-specific perturbation in mouse brain, from prep to tissue processing.
- Building a spatial transcriptomics toolset to model in vivo cell-type-specific genetic perturbations.

Department of Neurological Surgery, Columbia University, Graduate Researcher — New York, NY
intraoperative microneurography · DBS electrodes · human trigeminal ganglion · OR hardware design — 2024 – present

- Built the intraoperative recording setup for human trigeminal ganglion, using DBS electrodes.

Trends in Genomics, Columbia University, Teaching Assistant — New York, NY

- Co-designed the course for its first offering (2025). — Spring 2025 & 2026

Neural Engineering and Control Lab, Columbia University, Graduate Researcher (M.S.) — New York, NY
Neuropixels · GRAB sensors · 2-photon · optogenetics · chemogenetics · stereotaxic surgery · histology · behavior — 2018 – 2023

- Co-created neurOS, a Simulink Real-Time platform for multimodal closed-loop experiment control.
- Designed and implanted brain interfaces for Neuropixels, fiber photometry and 2-photon imaging.
- Developed the lab's in vivo perturbation assay and its cell-type-specific transcriptomic readout.
- Launched a longitudinal locus coeruleus stimulation study in an Alzheimer's model; basis of an R01 line.

Department of Biomedical Informatics, Columbia University, Student Researcher — New York, NY
retrospective case series · clinical data analysis — 2020

- Co-led 30 medical students and 18 clinicians characterizing the first COVID-19 wave in New York.

Professional History

MIT Lincoln Laboratory / MIT Haystack Observatory, Engineer, Advanced Sensors and Techniques Group — Westford, MA
radar systems · RF hardware · thermal design · field deployment · computer vision — 2016 – 2018

- Designed the structure and cooling for a solid-state radar transmitter, fielded in the Marshall Islands.
- Built a stand-alone traveling-wave-tube test facility; led the Millstone 84-foot azimuth encoder upgrade.
- Led a team adapting computer vision for autonomous UAV flight through an obstacle course.

MIT Lincoln Laboratory, Student Researcher, Systems Engineering Group — Lexington, MA
systems design · spacecraft propulsion · vacuum testing — 2015 – 2016

- Designed, built and characterized an experimental pico-satellite monopropellant thruster in vacuum.

GE Aviation / Unison Industries, Aerospace Engineering Intern — Jacksonville, FL
manufacturing engineering · design automation — 2014

- Automated the build instructions for jet-engine cable harnesses, generated from the design files.

Publications & Papers

Intraoperative microneurographic recording from the human trigeminal ganglion using standard deep brain stimulation electrodes
C. Slater, …, R. F. Sekula Jr.
Submitted (2026)

Intracranial delivery of neuron-preferential lipid nanoparticles for gene-editing activity in mouse brain
S. S. Cai, **C. Slater**†, …, K. W. Leong
Cell Reports Physical Science 7(9), 103514 (2026) [DOI] *† equal contribution*

Behavioral state-dependent norepinephrine dynamics in the primary somatosensory and prefrontal cortices during tactile detection tasks
C. Kelley, T. Lantin, **C. Slater**, …, Q. Wang
bioRxiv preprint (2026) [bioRxiv]

Alpha modulation of spiking activity across multiple brain regions in mice performing a tactile selective detection task
C. Kelley, **C. Slater**, …, Q. Wang
European Journal of Neuroscience 62(12), e70356 (2025) [DOI]

Bidirectional modulation of somatostatin-expressing interneurons in the basolateral amygdala reduces neuropathic pain perception in mice
A. Apte, J. Fernald, **C. Slater**, …, Q. Wang
Frontiers in Pain Research 6, 1602036 (2025) [DOI]

Brain-wide cell-type-specific noradrenergic modulation of the transcriptome
C. Slater, …, Q. Wang
bioRxiv preprint (2025) [bioRxiv]

The Neuromodulatory Role of the Noradrenergic and Cholinergic Systems and Their Interplay in Cognitive Functions: A Focused Review
C. Slater, …, Q. Wang
Brain Sciences 12(7), 890 (2022) [DOI]

Alzheimer's disease: An evolving understanding of noradrenergic involvement and the promising future of electroceutical therapies
C. Slater, Q. Wang
Clinical and Translational Medicine 11(4), e397 (2021) [DOI]

Characterising the long-term clinical outcomes of 1190 hospitalised patients with COVID-19 in New York City: a retrospective case series
S. M. Shoucri, …, **C. L. Slater**, …, J. E. Zucker
BMJ Open 11(6), e049488 (2021) [DOI]

Characterization and clinical course of 1000 patients with coronavirus disease 2019 in New York: retrospective case series
M. G. Argenziano, S. L. Bruce, **C. L. Slater**†, …, R. Chen
BMJ 369, m1996 (2020) [DOI] *† equal contribution*

Interventions to reduce short-wavelength ("blue") light exposure at night and their effects on sleep: A systematic review and meta-analysis
A. Shechter, K. A. Quispe, J. S. Mizhquiri Barbecho, **C. Slater**, L. Falzon
Sleep Advances 1(1), zpaa002 (2020) [DOI]

An Interprofessional Approach to Clinical Workflow Evaluation Focused on the Electronic Health Record Using Time Motion Study Methods
J. Schwartz, J. Elias, **C. Slater**, K. Cato, S. C. Rossetti
AMIA Annual Symposium Proceedings, 1187–1196 (2019)

Fellowships & Grants

Van C. Mow Fellowship — Sep 2021

National Institute on Aging T35AG044303 — Mar 2019
Brief Research in Aging and Interdisciplinary Neurosciences (BRAIN) Program

Presentations

Grounding RNA prediction models in structure and function — New York, NY, Oct 2026
Irving Institute for Cancer Dynamics Research Meeting, Columbia University. Speaker.

Whole-brain, cell-type-specific transcriptomic profiling of in vivo neurochemical perturbation — Chicago, IL, Oct 2024
Society for Neuroscience. Presenting author.

Leveraging large-scale electrophysiology array recording to identify neural biomarkers for improved closed-loop neuromodulation for chronic pain — Chicago, IL, Oct 2024
Society for Neuroscience. Contributing author.

Noradrenergic modulation of population activity in the primary somatosensory cortex, striatum, and amygdala in a tactile selective detection task — Chicago, IL, Oct 2024
Society for Neuroscience. Contributing author.

Locus coeruleus modulation of population activity in the awake somatosensory cortex — Washington, DC
Society for Neuroscience. Contributing author. — Nov 2023

Lipid nanoparticle-mediated gene editing in the mouse brain via intracranial delivery — Seattle, WA
Biomedical Engineering Society. Contributing author. — Oct 2023

Noradrenergic dynamics in the sensory and prefrontal cortices during a perceptual decision-making task — San Diego, CA
Society for Neuroscience. Presenting author. — Nov 2022

Functional noradrenergic regulation of the transcriptome during decision making under varying uncertainty — New York, NY
Center of Excellence in the Neuroscience of Decision-Making Kickoff, Columbia University. Presenting author. — Oct 2022

Development of a neural network architecture for optimal prediction of EEG alpha wave power — New York, NY
Northeast Bioengineering Conference, Columbia University. Presenting author. — Apr 2022

Targeted neural stimulation to improve neuropathic pain treatment — New York, NY
Northeast Bioengineering Conference, Columbia University. Presenting author. — Apr 2022

MIDA-based finite element modeling to optimize electrotherapy configurations — New York, NY
Northeast Bioengineering Conference, Columbia University. Contributing author. — Apr 2022

Long-term locus coeruleus stimulation reduces Aβ pathology in a mouse model of neurodegeneration — New York, NY
Medical School Research Symposium, Columbia University. Presenting author. — May 2021

Synthetic neuromodulation as a potential treatment for Alzheimer's disease — New York, NY
NIH Summer Research Symposium, Columbia University. Speaker. — Aug 2019

The temporal expression of consciousness — Worcester, MA
Class of 1879 Award Ceremony and Reception, WPI. Speaker. — Mar 2017

Leadership

President, Columbia Health Tech Assembly — Jun 2020 – Aug 2022

Senior Advisor, Innovative Medicine Club — Aug 2020 – Aug 2022

Student Member, VP&S Curriculum & Education Discover Subcommittee — 2020 – 2021

Ethics Representative, Medical School Class Council — Aug 2018 – Jul 2021

Treasurer, WPI Chapter of AIAA — Jan 2014 – Jan 2015

Honors & Awards

Best Session Poster, Medical School Research Symposium, Columbia University — May 2021

Roy and Diana Vagelos Scholarship — 2018 – 2021

D. & D. Merksamer Scholarship — 2018 – 2021

WPI Class of 1879 Prize for Outstanding Original Work in the Humanities and Arts — Mar 2017

WPI Graduation with Honors and High Distinction — May 2016

WPI Presidential Scholarship — 2012 – 2016

Most Impactful Project Award, WPI Interactive Qualifying Project — May 2015